



Natural language processing course

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Abstract

Large language models have difficulty providing accurate, up-to-date answers in rapidly evolving fields like competitive gaming. This project introduces a domain-specific AI assistant for League of Legends that uses Retrieval-Augmented Generation (RAG). The system indexes patch notes, champion statistics, and match history data and uses this information to retrieve relevant context at query time and generate informed, accurate responses. The assistant can answer questions about champion strengths and weaknesses and predict the viability of champion picks based on retrieved patch and matchup data. Evaluation combines automated retrieval metrics with human assessments of response quality and usefulness. The expected results will demonstrate that a domain-specific RAG reduces inaccurate responses and improves precision compared to a standalone LLM.

Keywords

Domain-Specific AI, Natural Language Processing, Retrieval-Augmented Generation, Game Analytics

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Introduction

In recent years, large language models (LLMs) have demonstrated impressive abilities in understanding and generating human-like text across various fields. However, despite their versatility, these models struggle with tasks requiring the latest specialized knowledge or a deep understanding of context.

These limitations are noticeable in competitive gaming environments, such as League of Legends. This complex multiplayer game has a continuously shifting strategic landscape due to frequent patch updates. The choices players make are based on a complex interaction of various factors, including the strengths and weaknesses of the champion, the items chosen, and past performance trends. This is something that a model based on statistics struggles to keep up with.

This work addresses these challenges by developing a domain-specific conversational assistant based on a RAG framework. Rather than relying on knowledge encoded during pretraining, the system dynamically retrieves relevant information from a tailored knowledge base containing patch notes, champion statistics, and match history data. This context is then incorporated directly into response generation. This design is particularly well-suited to League of Legends: when a new patch is released, only the knowledge base requires updating, not the model itself. The result is a system that can accurately answer nuanced questions about champion performance, recent balance changes, and matchup recommendations, surpassing the capabilities of standalone LLMs.

Related Work

Earlier machine learning studies in League of Legends have mostly concentrated on match outcome prediction and player performance assessment, attaining robust outcomes through gradient boosting, tensor factorization, and event-driven win probability models [1, 2, 3]. Jiang et al. [4] notably demonstrated that incorporating patch version context directly into predictive models improves accuracy. This confirms that game knowledge is inherently version-dependent and rapidly evolving.

However, these models primarily operate on structured data and lack mechanisms for interactive knowledge access. This has led to growing interest in NLP approaches.

Lee et al. [5] introduced a voice-based large language model companion for beginner League of Legends players in their 2026 study, LeagueBot, reporting significant reductions in cognitive challenge and perceived tension in a controlled experiment with 33 players. However, participants reported receiving recommendations for nonexistent items. The authors attributed this hallucination problem to the LLM's lack of access to current game data. RAG was explicitly proposed as a solution, but latency concerns in fast-paced gameplay prevented its implementation. Motivated by these limitations, this work presents an RAG-based assistant for League of Legends that retrieves current patch notes, champion statistics, and match data, making it possible to provide practical, up-to-date responses.

Data

We leverage the official Riot API[6] alongside player data sourced from OP.GG[7] as our data. Beyond in-game performance metrics, we integrate official documentation and community knowledge to guide feature engineering and model training. Additionally, we utilize balance patch releases from Riot Games to capture and account for evolving gameplay mechanics across different patch versions.

Game Rules and Domain Knowledge

The foundational layer of the knowledge base consists of official and community-maintained documentation describing the rules, objectives, and mechanics of League of Legends. These sources provide structured, stable textual descriptions of core concepts such as champion roles, map objectives, itemisation, and win conditions. Since this content changes infrequently, it serves as a static background knowledge layer, ensuring consistent understanding of game fundamentals.

Beginner and Strategy Guides

To ensure the assistant can respond to practical, player oriented queries in natural language, beginner and strategy guides are incorporated. These documents reflect how real players formulate questions and what useful answers look like in practice. This category also informs the construction of the evaluation question set, as the phrasing and topics closely mirror the queries the assistant is expected to handle.

Patch Notes

Patch notes represent the primary and most critical component of the RAG knowledge base. Riot Games releases balance updates approximately every two weeks, introducing champion buffs, nerfs, item changes, and systemic adjustments that fundamentally alter the competitive landscape. Each patch document is chunked, embedded, and indexed into the vector store, allowing the retrieval pipeline to surface the most relevant balance changes in response to user queries. This design enables the knowledge base to be updated with each new patch without any modification to the underlying model.

Champion Statistics and Meta Data

Champion performance statistics are computed from match data collected via the Riot API, calculating win rates, pick and ban rates, optimal item builds on a per-patch and per-rank basis. This data is inherently dynamic. A champion considered strong in one patch may be significantly weaker following a targeted nerf, making it well-suited for RAG retrieval. When a user asks whether a specific champion is viable in the current meta, the retrieval pipeline surfaces the most recent computed statistics, grounding responses in empirical data derived from observed gameplay.

Match Data

Historical match data is obtained via the Riot Games Developer API, which provide granular per-game information

including champion selections, item builds, kill events, gold progression, and final outcomes. This data serves primarily as ground truth for evaluation: champion pick recommendations can be assessed against actual match outcomes from Diamond+ tier games to quantify predictive accuracy.

References

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